

Using Finite Wing Analysis to Select the Optimal Small-aircraft Wing Design

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Nomenclature

$\vec{A}$	=	vector of Fourier coefficients
A_n	=	Fourier coefficient n
a_0	=	slope of the lift coefficient vs angle of attack plot for an airfoil
AR	=	aspect ratio
b	=	length of the span of the wing
$\vec{b}$	=	difference between geometric angle of attack and zero lift angle of attack at each discretization point
c	=	chord length
$C_{D,i}$	=	coefficient of induced drag
c_l	=	lift coefficient per unit span
D'_i	=	induced drag force per unit span
D_i	=	induced drag force
e	=	Oswald's Efficiency Factor or span efficiency factor
j, k, l, m, n	=	counter variables
L'	=	Lift force per unit span
L	=	total lift force
M	=	discretization point along the span of the wing
N	=	number of terms in Fourier series
S	=	wing reference area
V_∞	=	freestream flow velocity

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w	=	downwash velocity
y	=	position along the a bound vortex
y_0	=	a particular point along the span of the wing
α	=	angle of attack
α_{eff}	=	effective angle of attack
α_i	=	induced angle of attack
$\alpha_{L=0}$	=	zero-lift angle of attack
Γ	=	circulation
Γ_0	=	maximum circulation along the span for an elliptic distribution
δ	=	used to define Oswald's Efficiency Factor
θ	=	parameterized position along wing span
θ_0	=	a particular point along the span of the wing in the parameterized coordinate
ρ	=	freestream density of flow

I. Introduction

Aerospace wing design plays an important role in the development of any winged aircraft, since the form the wing takes dictates the performance characteristics of the aircraft. Early aerodynamicist determined that an elliptic wing planform has the greatest efficiency from the standpoint of induced drag. This investigation explores the validity and limitations of this finding, and compares an elliptic wing planform with other more simple planforms. The real world problem compares key characteristics of these wing shapes, the historical evolution and modern usage of these wing shapes, and common performance metrics by which the performance of these wings is evaluated. The academic-style problem delves into a computational investigation of wing performance. A mathematical derivation motivates the performance metrics and characteristics of interest, and the code builds off of this mathematical basis to model, predict, and compare the performance of different wing shapes to develop a recommendation for a realistic set of design requirements.

The topics explored in this report directly build off of core course concepts. Prandtl's lifting line theory undergirds the academic investigation. This theory directly builds off of many course concepts, including the Kutta-Joukowski theorem for circulation and lift, lift and drag relationships with angle of attack, and drag polars. Induced drag, wingtip vortices, and downwash cause and define many of the differences in performance of the wing shapes investigated. Even wing shape, specifically the span efficiency of elliptic wings, was discussed to some extent in lecture. The idea, discussed in lecture, that a tapered wing could reasonably approximate an elliptic wing planform in terms of efficiency largely motivated us to pursue this investigation. By analyzing elliptic, rectangular, and tapered wings both theoretically and

numerically, this report not only contextualizes course concepts but also models the type of preliminary aerodynamic analysis performed in professional aerospace design.

II. Real-World Problem

A. Historical Context

While the first successful powered flight was made by the Wright brothers, there was an aviator who came before them. Otto Lilienthal was a German engineer that was a pioneer in unpowered gliders. His first research on flight was studying birds as a child and a teenager. After completing his education, he and his brother started developing different gliders [1]. In 1867 they developed their first glider, which was unsuccessful. He continued his research in aviation, and in 1891 he created his first successful glider. He named this glider the Derwitzer Glider model. This glider, which is shown in figure 1, was made of willow rods and cotton fabric. Using his body weight and moving his legs to control the glider, he was able to glide 80 feet on his first successful flight [1].



Fig. 1 Image of Otto Lilienthal flying in one of his gliders [2]

The Wright brothers were inspired by Lilienthal's gliders and research. They based the wings of their original plane on his research. Through Lilienthal's studies and their own research, they decided that higher aspect ratios were better for flight. The aspect ratio of a wing is the length of the span compared to the chord. The span is the distance from one wingtip to another and is perpendicular to the fuselage. The chord of a wing is the length from the leading edge, which is at the front of the wing, to the trailing edge at the back of the wing. Their first successful glider had an aspect ratio of 6.7. The Wright Flyer, which was the first propeller-powered aircraft, had an aspect ratio of 6.4 [2]. This aspect ratio

is very similar to many aircraft that are still used today. Another revolutionary feature of the Wright Flyer were they wingtips of the airplane. The wingtips were able to warp in either direction, which was the controlling mechanism to roll the airplane. This was the first type of control surface and was revolutionary leading to the creation of ailerons [2]. The Wright brothers used a rectangular planform for their original glider. A planform is the shape of the wing when looking at it from above. The rectangular planar was most commonly used for all aircraft until 1925 when the first elliptical planform was first developed. In 1918 theorist Ludwig Prandtl developed his theory for finite wing span. In this study, it was shown that the way to produce maximum lift was to make an elliptical lift distribution. Further in the study, it is determined that the best way to produce an elliptical lift distribution is through an elliptical wing planform. In 1925 the first elliptical winged airplane took off. This airplane was called B umer Sausewind and was made by a company of the same name [3]. The production of these airplanes was the start of modern planform studies.

B. Elliptic Planform

Ludwig Prandtl was a German physicist who studied the aerodynamic effects on different kinds of wings. Through his research, Prandtl found that an elliptical lift distribution leads to a maximum possible span efficiency factor. The span efficiency factor for a wing dictates the amount of lift a wing can have with minimum drag. Prandtl's data on elliptical lift were then used to create the first elliptical planform airplane, the B umer Sausewind. This plane was a small sports plane that was created by a company of the same name B umer Sausewind [3]. This led to the development of many more elliptical-shaped airplanes. The elliptical shape of the wing leads to a higher coefficient of lift and a lower induced drag. This allowed these airplanes to reach very high top speeds at the time and was especially used in World War 2. One of the most popular planes with the elliptical-style planform was the Supermarine Spitfire. The Spitfire shown in Figure 2 was a very popular British fighter plane used in World War 2. The Spitfire's elliptical wing design was influenced by aerodynamic theories from Prandtl. The combination of the elliptical planform and the thinness of the airfoil allows the Spitfire to have very low induced drag. This allows the Spitfire to reach high speeds and heights with reduced induced drag. The Spitfire also had excellent maneuverability, which made it very efficient in dogfights [4].



Fig. 2 Image of the Submarine Spitfire [5]

Although at the time elliptical wings were revolutionary with the improvement of modern technology and simulations, we now know that elliptical wings have a lot of flaws. The biggest flaw with elliptical wings is the instability and stall characteristics. Elliptical wings tend to have more of an abrupt stall behavior. The reason for this is because the angle of attack and lift coefficient are nearly uniform, which causes the entire wing to stall at once. This is much less safe to fly and causes many concerns among pilots. Other planform shapes are much safer and offer much more gradual and predictable stall characteristics [6]. Another problem with the elliptical wing is the complexity of manufacturing. Compared to other planar wing types such as rectangular, the manufacturing of rounder surfaces is much more difficult. Another issue with the manufacture of elliptical airfoils is maintaining structural integrity. The load distribution on elliptical wings can cause the need to add structural reinforcements, which can add weight to the plane and reduce aerodynamic performance [6]. With all of these issues, elliptical wings are not used in many modern aircraft.

C. Rectangular Planform

Rectangular planforms were the first types of wing shape, even being used by the Wright brothers on their original Wright Flyer airplane. Rectangular planforms can be identified by their wing shape, where the length of the chord remains constant from the root to the tip. The length of the chord is the distance from the leading edge to the trailing edge. The root of the wing is the part of the wing attached to the fuselage and the tip is the opposite side of the wing. This geometry makes the wing look like a rectangle when viewed from above. Rectangular wings are very common among aircraft that fly at lower speeds. This is due to the simplicity of the manufacturing of these wings. These wings also have very predictable stall behavior; this is because rectangular wings are more likely to stall at the root before the tip. This allows the pilot to still be able to control the plane while in stall. For this reason, rectangular wings are commonly used for training aircraft[7].



Fig. 3 Image of Cessna 172-R Skyhawk [8]

The image shown above in Figure 3 is a Cessna 172-R Skyhawk. The Cessna Skyhawk is an American made 4 seated single engine aircraft. It was first flown in 1955, more Skyhawks have been produced than any other aircraft [9]. The Skyhawk has been used as a training aircraft for decades. This is due to its high lift and low speeds and stability. The Skyhawk has a mostly rectangular-shaped planform and that is a cause of these preferable aerodynamical features. With an aspect ratio of 7.32 this allows the Skyhawk to be able to produce a great amount of lift with minimal induced drag. This aspect ratio also allows the wings to be structurally sound and without the need for excess supports. The benefit of this is that it reduces the overall weight of the plane. The Skyhawk uses a single wing that is in a higher position. This will increase its aerodynamic performance and its stability and operation requirements [9]. In general, the Cessna Skyhawk is a great model of the benefits of using a rectangular planform for wing design. Shown in Figures 4 and 5 are the Cessna Skyhawk lift and drag coefficients for different angles of attack.

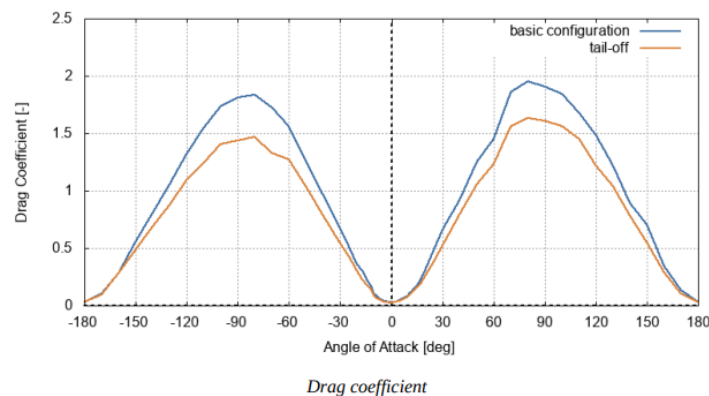


Fig. 4 Plot of the drag coefficient versus the angle of attack for a Cessna 172-R Skyhawk[7]

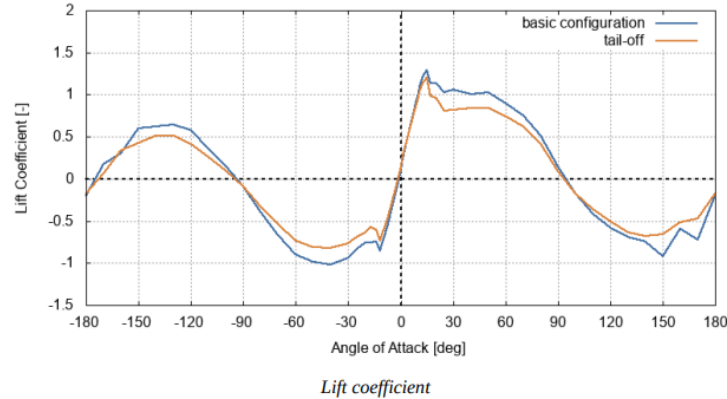


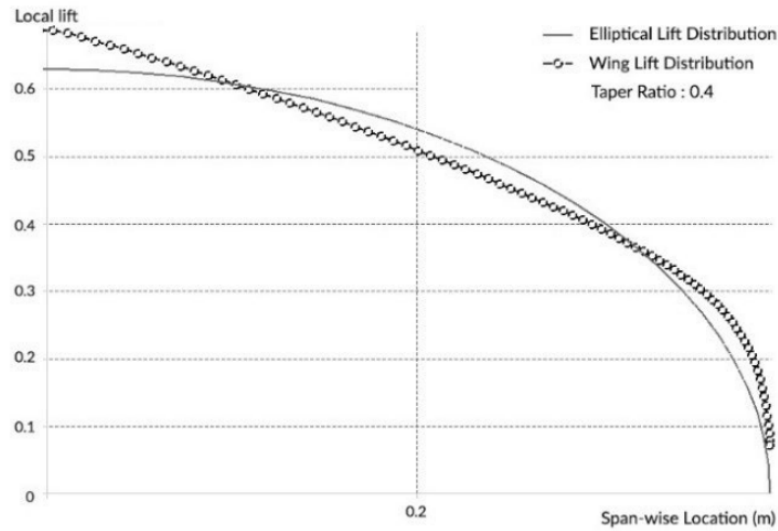
Fig. 5 Plot of the drag coefficient versus the angle of attack for a Cessna 172-R Skyhawk[7]

These figures show the lift and drag coefficients for the Skyhawk with different configurations. The orange line is the lift and drag with only the main wing. The blue line is the lift and drag with the main wing and the tail wings. In Figure 4 the drag coefficient for the Cessna increases with respect to the angle of attack. The increase in angle of attack leads to an increase in both induced drag and parasitic drag. Induced drag increases because as the angle of attack increases, the lift increases as well. With an increase in lift, the induced drag will increase as a result of the increase in vortices at the wingtips. Parasitic drag is the drag caused by the drag of the geometric shape of the airfoil and the pressure differences as the plane moves through the air. The higher the coefficient of lift the greater the geometric area and the leading edge which increases pressure difference. Figure 5 shows the coefficient of lift for different angles of attack. The maximum coefficient of lift is around 15 degrees for the angle of attack. At this angle, the plane produces the most lift with a lift coefficient of 1.25. This angle of attack corresponds to a drag coefficient of 0.2. This coefficient of lift compared to the drag coefficient is the reason why rectangular planforms are pretty efficient. At low speeds, they produce high amounts of lift but with moderate amounts of drag. The Cessna Skyhawk is a perfect example of the type of plane in which rectangular planforms are ideal. It is able to produce high lift at slower speeds, and while it has a moderate amount of drag, it is more stable and easy to control [7].

D. Tapered Planform

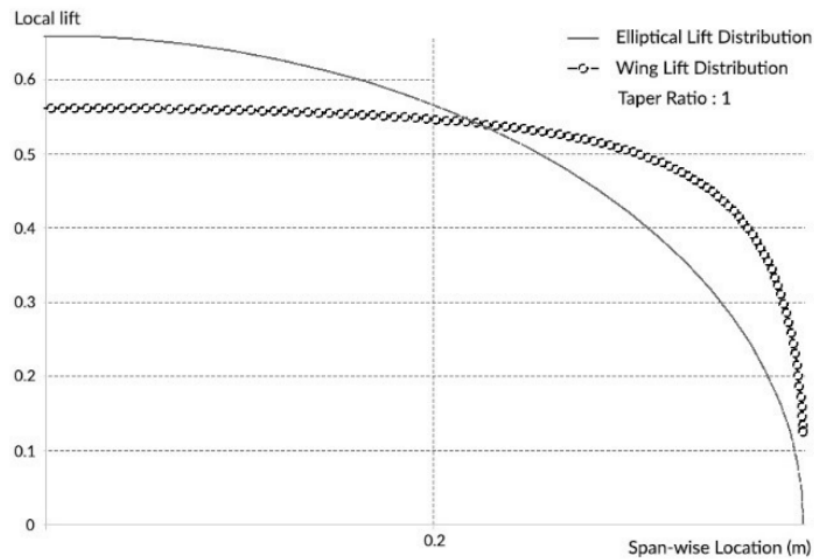
While rectangular planforms have many benefits that are many issues with their aerodynamics specifically with the amount of drag produced. A reason for this is the amount of surface area at the tip of the wing. This higher surface area leads to a higher moment for the wing, which requires more structural support. Another issue with the tip of the wing is that with a higher surface area, surface drag is higher. This leads to poorer aerodynamic performance and room for improvement. The solution to this is to taper the wings. Tapered wings have a slightly different shape than rectangular wings. Instead of the length of the span being equal to the length of the chord, the chord changes length throughout the span of the wing. The taper ratio of a wing is the length of the tip divided by the length of the root. Usually, the length

of the chord at the tip is less than the length at the root. This allows the surface area of the wing compared to the chord to vary along the span. The lower surface area near the tip is ideal because it allows the wing to have a more desirable lift distribution [10]. According to Prandtl lifting line theory, a maximum amount of lift is produced in an elliptical lift distribution. A study published by the European Mechanical Science group confirms this claim. They tested different taper ratios among a span-wise location by looking at the plotting the local lift at each location. They then compared these data with what an elliptical distribution would look like [10].



(b)

Fig. 6 Local lift versus spanwise location for an elliptical wing and a wing with a taper ratio of 0.4 [10]



e)

Fig. 7 Local lift versus spanwise location for an elliptical wing and a rectangular wing [10]

The plots in Figures 6 and 7 show the immediate benefits of tapered wings compared to rectangular wings. The solid line is the plot of the elliptical lift distribution which is the maximum lift a wing can produce. The line with circles throughout is the lift of the tapered wing. Figure 6 is the plot of the elliptical lift distribution compared to a wing with a taper ratio of 0.4. This taper ratio provides the maximum lift distribution and, as seen in the plot, is closest to the elliptical lift distribution. Figure 7 shows the same elliptical lift distribution, but plotted against a wing with a taper ratio of one. A taper ratio of one means that the length of the tip is equal to the length of the root. This plot shows that the lift produced by the rectangular wing is not even close to the lift produced by the elliptical wing. In addition to lift, tapered wings also produced significantly less drag than rectangular wings [10].

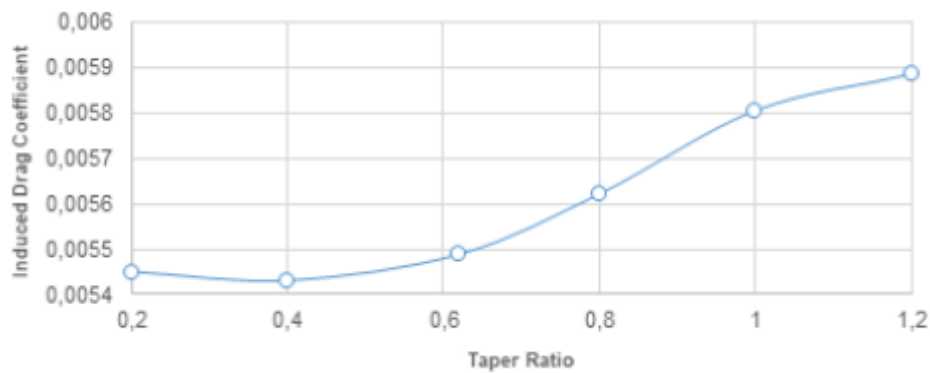


Fig. 8 Taper ratio compared to the induced drag for a tapered wing [10]

This plot shows how the different taper ratios change the induced drag felt by the airplane. The plot shows that the lowest drag ratio is felt when there is a 0.4 taper ratio. This would be ideal when designing the wings of an airplane. Taper planforms are much better at producing lift and minimizing drag than rectangular planforms. This is because when tapering the tip part of the wing is tapered it forms a closer lift distribution to an elliptical-like lift distribution, which is more efficient [10].

E. Efficiency of Wings

While it is important to select a planform that will be most efficient for your plane, there are other ways to make your plane more efficient. One way to measure the efficiency of a wing can be defined as the amount of lift that it can produce compared to the amount of drag it produces. One way to determine this efficiency of a wing is to use the Oswald efficiency factor. The Oswald factor takes key geometrical parameters of a wing and calculates a value. This value defines how efficient a wing is in terms of aerodynamics and fuel consumption. The geometric characteristics used for the Oswald factor are the aspect ratio and the taper ratio, the sweep angle, the wing area, and the fuselage interference [11]. The sweep angle is defined as the angle between the leading edge and the line perpendicular to the longitudinal axis of the aircraft. The fuselage interference is the aerodynamical interaction between the wing and the

fuselage. The sweep angle and the fuselage interference are very important factors in wing efficiency but can not be modeled by Prandtl Lifting Line theory which is used in the academic problem. The Oswald factor is also designed for only subsonic aircraft configurations but is still very effective for determining the effectiveness of a wing [12].

F. Conclusion

When selecting the type of planform and different wing characteristics, it is important to consider the type of aerodynamic qualities that would be ideal. When looking at the rectangular planform, it can produce high amounts of lift at lower angles of attack while still having high stability. The problem with this type of wing is the high amount of drag it produces. The ellipse wing, while ideal in its lifting quality, is much more likely to stall and much harder to produce. The tapered wing is a combination of both of those types of wings and is frequently used in many of today's airplanes. The tapered-winged airplane can produce high amounts of lift while minimizing drag. It also has high stability and is easily manufacturable.

G. Connections To Class

The evaluation of finite wings and defining their aeronautical characteristics is very relevant for what is discussed in incompressible flow. In class we have discussed concepts like induced drag, Prandtl Lifting Line theory, and the analysis of different aspect ratios. All of these concepts are relevant for selecting a wing planform for a specific airplane. To be able to design a wing, one needs to be able to understand all the geometric factors that will affect the wing aerodynamically. This is critical for any aerodynamic project.

III. Explanation of Academic-Style Problem

A. Problem Statement

You are hired to consult for a small aerospace startup. They are looking for a way to run preliminary calculations on wing design to help inform their early product life cycle design choices. Their first plane design shall be a small single-passenger recreational aircraft. Their aircraft must be designed to have a maximum wingspan of 12 meters, so it can fit in smaller airport hangars. Given that the startup is small, wings with elliptic planforms are not in their manufacturing budget. Use finite-wing theory to help the company choose the proper design.

B. Assumptions

As with any analytical model, assumptions must be made to make real-life phenomena feasibly calculable. Here are the assumptions made for the academic problem:

- The effect of all other components of the plane on the wing are ignored.
- The flow shall be incompressible.

- The flow over the wing shall be steady.
- The flow over the wing shall be inviscid.
- The flow over the wing shall be irrotational.
- Only straight (non-swept) wings with no dihedral angle shall be considered.
- Small angle approximations for angle of attack are made during the Methods section of this report.
- Flow remains attached along the entire wingspan.
- The wing shall not have aerodynamic twist.

C. Methods

1. Effective Angle of Attack

Finite wings generate a downwash, causing a local downward flow behind the wing within its wingtips. This downward component of velocity acts as a downward component of the freestream velocity, resulting in the lift vector on the wing to tilt backward to remain perpendicular to the local relative flow. This new backward component of lift opposes the motion of the wing, causing induced drag. The amount that the free stream velocity is displaced downward by downwash is called the induced angle of attack.

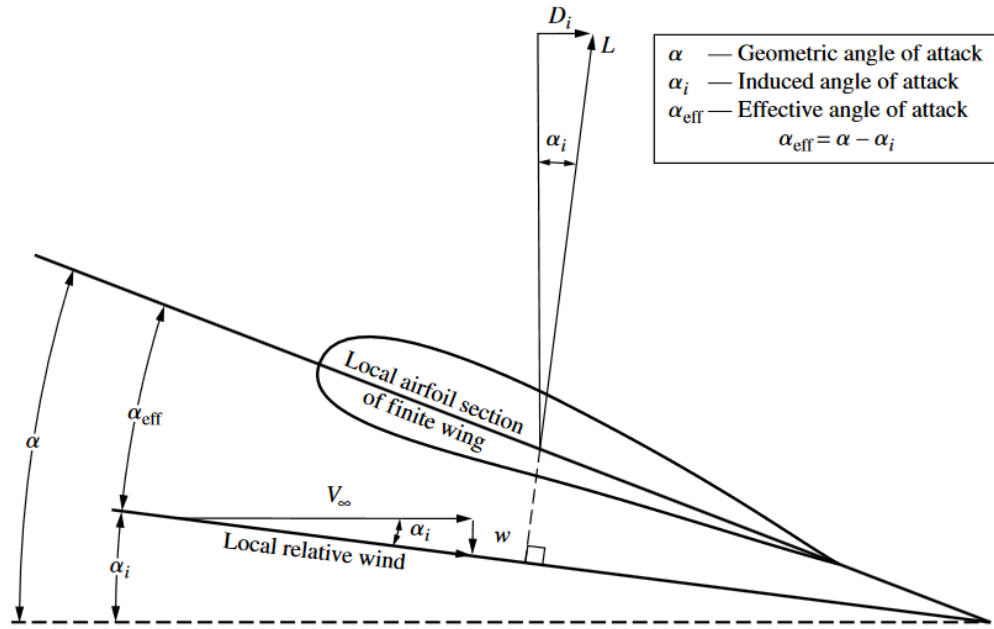


Fig. 9 Effect of downwash on the local flow over a local airfoil section of a finite wing [13].

$$\alpha = \alpha_{\text{eff}} + \alpha_i \quad (1)$$

While simple, this equation is the basis of the fundamental equation of Prandtl Lifting Line theory. The effective angle of attack is the angle of attack that the wing actually "sees" while moving through the air. Given the positive correlation (and direct proportionality for symmetric airfoils in Thin Airfoil Theory) between coefficient of lift and angle of attack, the loss of lift due to induced drag is clearly illustrated by the equation, since the downwash subtracts from the geometric angle of attack relative to the horizontal.

2. The Vortex Filament and Prandtl Lifting-Line Theory

The vortex filament is the three-dimensional analog to the point vortex in two-dimensional potential flow analysis. The vortex filament extends infinitely along its axis and has a constant strength of circulation distribution. Like point vortices, vortex filaments can be leveraged to simulate lifting bodies, specifically finite wings. This is the basis of Prandtl's Lifting-Line Theory.

Hermann Helmholtz stated that a vortex filament can not end in a fluid. It must either extend infinitely or form a closed path. Keeping this in mind, Prandtl harnessed vortex filaments by simulating finite wings with horseshoe vortices. Horseshoe vortices consist of a line segment called a bound vortex that simulates wing lift generation and two semi-infinite filaments trailing off the tips of the bound vortex to simulate trailing vortices on wings. Hence, the horseshoe vortex modeled the most important considerations of three-dimensional, finite wings. The downwash due to a vortex filament at a point y along the bound vortex can be calculated using the Biot-Savart Law:

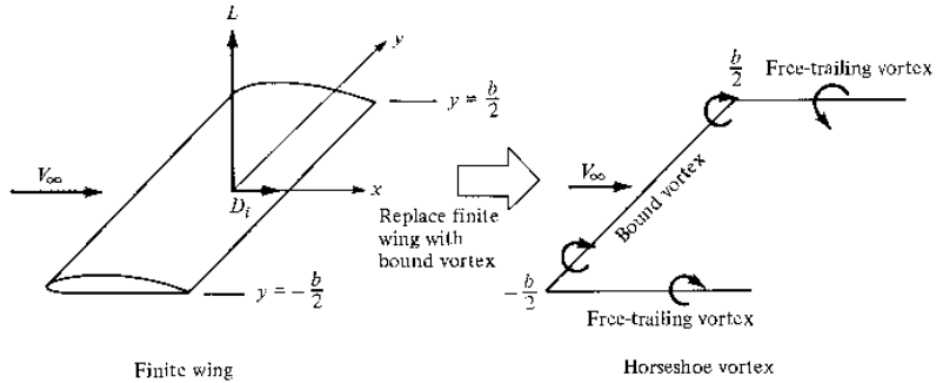


Fig. 10 Using a Horseshoe Vortex to Model a Finite Wing [13].

$$w(y) = -\frac{\Gamma}{4\pi(\frac{b}{2} + y)} - \frac{\Gamma}{4\pi(\frac{b}{2} - y)}$$

$$\text{Reduces} \rightarrow w(y) = -\frac{\Gamma}{4\pi} \frac{b}{\left(\frac{b}{2}\right)^2 - y^2} \quad (2)$$

However, a single vortex filament breaks down at the endpoints of the bound vortex, approaching an infinitely negative value for downwash. To remedy this, Prandtl determined that a wing should not be modeled by one horseshoe

$$\alpha_i(y_0) = \tan^{-1} \left(\frac{w(y_0)}{V_\infty} \right)$$

Efficient of lift can be related to angle of attack: $C_L = C_{L_0} + C_{L_1}(\alpha) + C_{L_2}(\alpha^2)$

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$$L' = \frac{1}{2}\rho V_\infty^2 c(y_0) c_l = \rho V_\infty \Gamma(y_0) \rightarrow c_l = \frac{2\Gamma(y_0)}{V_\infty c(y_0)} \quad (7)$$

$$(6) \text{ and } (7): \quad \frac{2\Gamma(y_0)}{V_\infty c(y_0)} = 2\pi [\alpha_{\text{eff}}(y_0) - \alpha_{L=0}]$$

$$\alpha_{\text{eff}}(y_0) = \frac{\Gamma(y_0)}{V_\infty c(y_0)} + \alpha_{L=0}(y_0) \quad (8)$$

$$\alpha(y_0) = \frac{\Gamma(y_0)}{\pi V_\infty} + \alpha_{L=0}(y_0) + \frac{1}{4\pi V_\infty} \int_{-b/2}^{b/2} \frac{(d\Gamma/dy)dy}{y_0 - y} \quad (9)$$

Equation (9) is known as the Fundamental Equation of Prandtl's Lifting-Line Theory. Upon closer inspection, it is just a restatement of equation (1): The geometric angle of attack is equal to the sum of the effective angle of attack and the induced angle of attack. However, equation (8) was substituted in for the effective angle of attack, and equation (5) was substituted in for the induced angle of attack. Note in this equation that the only unknowns for a given point along the span of a given finite wing are the circulation and its derivative. This fact shall be elaborated on later in this derivation.

To calculate total lift for a finite wing, the lift per unit span must be integrated over the span of the finite span. Again, the Kutta-Joukowski Theorem is utilized for the lift per unit span:

$$L = \int_{-b/2}^{b/2} L'(y) dy = \rho_\infty V_\infty \int_{-b/2}^{b/2} \Gamma(y) dy \quad (10)$$

Once again, the relation between lift and lift coefficient can be used to find the lift coefficient:

$$C_L = \frac{L}{\frac{1}{2}\rho_\infty V_\infty^2 S} = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy \quad (11)$$

A similar process is used to calculate induced drag and induced drag coefficient. Using Fig. 9 and trigonometry, the drag per unit span is:

$$D'_i = L'_i \sin(\alpha_i)$$

Again making use of small angle approximations:

$$D'_i = L'_i \alpha_i \quad (12)$$

Integrating equation (12):

$$D_i = \int_{-b/2}^{b/2} L'(y) \alpha_i(y) dy = \rho_\infty V_\infty \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y) dy \quad (13)$$

Relating induced drag coefficient to induced drag given in equation (13):

$$C_{D,i} = \frac{D_i}{\frac{1}{2} \rho_\infty V_\infty^2 S} = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y) dy \quad (14)$$

Equations (9), (11), and (14) demonstrate the importance of circulation in Prandtl Lifting-Line Theory. Solving for circulation lets one calculate the most important parameters for wing design, such as coefficient of lift and induced drag coefficient. The rest of the values in equations (9), (11), and (14) are constants in the problem or can be determined by the characteristics of the wing at a given point along the span. Therefore, solving for circulation is the motivation for the remainder of the derivation.

3. Elliptic Wing Distribution

The elliptic circulation distribution was the first approach for evaluating lift, induced drag, downwash, etc. for finite wings. Rather than trying to calculate the circulation in equation (9), he proposed the following circulation distribution:

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{2y}{b}\right)^2} \quad (15)$$

Note that this circulation distribution reaches a maximum in the center of the wing and decreases elliptically with y along the span; the circulation at the wing tips is zero. This circulation distribution is called the elliptic circulation distribution.

Given the Academic-Style problem's statement that elliptic wing distributions shall not be considered due to their manufacturing complexity, the derivation shall not focus on the full derivation for Prandtl Lifting Line Theory with an elliptic wing distribution. However, equations are developed that are vital to later calculations for general wing distributions. Therefore, the necessary equations will be provided, but steps in the derivation will be skipped.

Important relations between induced angle of attack, coefficient of lift, and induced drag coefficient are generated

from the elliptic wing distribution:

$$\alpha_i = \frac{C_L}{\pi AR} \quad (16)$$

Where:

$$AR \equiv \frac{b^2}{S}$$

A relation between induced drag coefficient and lift coefficient is made by using equations (14) and (16) and performing some simplifications to yield:

$$C_{D,i} = \frac{C_L^2}{\pi AR} \quad (17)$$

4. General Wing Distribution

To examine other planform shapes, there must be a way to take the results from the elliptic wing distribution to other planform shapes. It starts with the following transformation:

$$\text{Change of variables: } y = -\frac{b}{2} \cos \theta \quad (18)$$

$$\text{For an elliptic lift distribution, } \Gamma = \Gamma_0 \sin \theta \quad (19)$$

Given the above elliptic lift distribution, the natural next step for generalization is to use a Fourier Series. A Fourier Series is a mathematical technique that can recreate any periodic signal by adding together co-sinusoid functions of different frequencies, each scaled by its own Fourier coefficient. Hence, the general circulation distribution using the Fourier Series is shown below:

$$\Gamma(\theta) = 2bV_\infty \sum_1^N A_n \sin(n\theta) \quad (20)$$

Note that $2bV_\infty$ is pulled out of the general distribution Fourier Series. This is done to ensure that the Fourier coefficients (A_n) are dimensionless values. This makes it easier to scale the results general wing calculations for different

wing shapes and sizes. Next, the derivative of equation (20) is taken:

$$\frac{d\Gamma}{dy} = \frac{d\Gamma}{d\theta} \frac{d\theta}{dy} = 2bV_\infty \sum_1^N nA_n \cos(n\theta) \frac{d\theta}{dy} \quad (21)$$

Substituting equations (20) and (21) into (9) yields:

$$\alpha(\theta_0) = \frac{2b}{\pi c(\theta_0)} \sum_1^N A_n \sin(n\theta) + \alpha_{L=0}(\theta_0) + \sum_1^N nA_n \frac{\sin(n\theta_0)}{\sin(\theta_0)} \quad (22)$$

Where $c(\theta_0)$ is chosen based on wing geometry, and $\alpha_{L=0}(\theta_0)$ depends on the characteristics of the airfoil. Equation (22) is the general wing distribution form of the fundamental equation of Prandtl Lifting Line Theory. Now, instead of Γ and its derivative being the unknowns in this equation, the only unknowns are the Fourier coefficients. However, equation (22) can be reconfigured into N equations for N unknowns. This creates a solvable system of equations, allowing for the lift distribution to be calculated for a general wing planform.

The steps for creating a matrix system of equations are shown below.

$$\text{Set up discretization points:} \quad \theta_j = \frac{j\pi}{N+1} \quad j = 1, 2, 3, \dots, N$$

These discretization points represent physical points along the span of the wing. As mentioned before, the angle of attack, zero lift angle of attack, chord length are known and can be substituted into the equation. As the number of discretization points increases, the more faithful the calculated lift will be to the theoretical continuous lift distribution. This is because you are sampling more and more points along the span to improve accuracy. In the limit where infinite discretization points are considered, the circulation distribution would be the theoretical continuous lift distribution.

Also, note that the endpoints of the wing are not considered. This is because the system of equations will break down due to a singularity with the $\sin \theta_j$ term in the denominator, which will become apparent in the next step. As a result, the circulation at the endpoints is fixed at zero.

$$\text{At each point:} \quad \alpha(\theta_j) - \alpha_{L=0}(\theta_j) = \sum_1^N \left(\frac{2b}{\pi c(\theta_j)} \sin(n\theta_j) + n \frac{\sin(n\theta_j)}{\sin(\theta_j)} \right) A_n$$

Therefore the system can be represented as: $M\vec{A} = \vec{b}$

Where M can be represented in matrix form as:

$$\begin{pmatrix} M_{1,1} & M_{1,2} & M_{1,3} & \cdots & M_{1,N} \\ M_{2,1} & M_{2,2} & M_{2,3} & & \\ M_{3,1} & M_{3,2} & M_{3,3} & & \\ \vdots & & & \ddots & \\ M_{N,1} & & & & M_{N,N} \end{pmatrix}$$

With:

$$M_{l,m} = \frac{2b}{\pi c(\theta_l)} \sin(m\theta_l) + m \frac{\sin(m\theta_l)}{\sin(\theta_l)}$$

$M_{l,m}$ represents some an entry in the matrix M in the l th row and m th column, with $l \leq N$ and $m \leq N$. l denotes that the equation is being evaluated at the l th discretization point, and m denotes that the equation is being evaluated for the m th Fourier coefficient. More generally, the each in matrix M represents a discretization point along the span of the wing, while each column represents a Fourier coefficient in the Fourier Series.

$\vec{A}$ represents the vector of Fourier coefficients of length N . $\vec{b}$ represents the difference between the geometric angle of attack and zero lift angle of attack at each discretization point along the span. Like $\vec{A}$, $\vec{b}$ also has a length of N .

Here is the full matrix-vector system:

$$\begin{pmatrix} M_{1,1} & M_{1,2} & M_{1,3} & \cdots & M_{1,N} \\ M_{2,1} & M_{2,2} & M_{2,3} & & \\ M_{3,1} & M_{3,2} & M_{3,3} & & \\ \vdots & & & \ddots & \\ M_{N,1} & & & & M_{N,N} \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ \vdots \\ A_N \end{pmatrix} = \begin{pmatrix} \alpha(\theta_1) - \alpha_{L=0}(\theta_1) \\ \alpha(\theta_2) - \alpha_{L=0}(\theta_2) \\ \alpha(\theta_3) - \alpha_{L=0}(\theta_3) \\ \vdots \\ \alpha(\theta_N) - \alpha_{L=0}(\theta_N) \end{pmatrix}$$

As mentioned before, the equations for lift coefficient and induced drag coefficient are used to obtain the corresponding formulas for a general wing distribution. First is the lift coefficient:

$$C_L = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) dy = \frac{2b^2}{S} \sum_1^N A_n \int_0^\pi \sin(n\theta) \sin \theta d\theta$$

$$\int_0^\pi \sin(n\theta) \sin \theta d\theta = \begin{cases} \frac{\pi}{2} & n = 1 \\ 0 & n \neq 1 \end{cases}$$

Therefore, the coefficient of lift only depends on the first Fourier coefficient: (23)

$$C_L = A_1 \pi \frac{b^2}{S} = A_1 \pi AR \quad (24)$$

A similar procedure is used for induced drag coefficient:

$$C_{D,i} = \frac{2}{V_\infty S} \int_{-b/2}^{b/2} \Gamma(y) \alpha_i(y) dy = \frac{2b^2}{S} \int_0^\pi \sum_1^N A_n \sin(n\theta) \alpha_i(\theta) \sin \theta d\theta \quad (25)$$

$$\alpha_i(\theta_0) = \frac{1}{\pi} \sum_1^N n A_n \int_0^\pi \frac{\cos(n\theta)}{\cos \theta - \cos \theta_0} d\theta$$

$$\rightarrow \alpha_i(\theta_0) = \sum_1^N n A_n \frac{\sin(n\theta_0)}{\sin \theta_0} \quad (26)$$

$$(26) \text{ into } (25) \rightarrow C_{D,i} = \frac{2b^2}{S} \int_0^\pi \left(\sum_1^N A_n \sin(n\theta) \right) \left(\sum_1^N n A_n \sin(n\theta) \right) d\theta \quad (27)$$

$$\int_0^\pi \sin(m\theta) \sin(k\theta) d\theta = \begin{cases} \frac{\pi}{2} & m \neq k \\ 0 & m = k \end{cases}$$

Hence, the mixed terms all become zero, so (27) becomes:

$$C_{D,i} = \pi AR \sum_1^N n A_n^2 = \pi AR (A_1^2 + \sum_2^N n A_n^2)$$

$$C_{D,i} = AR \pi A_1^2 \left(1 + \sum_2^N n \left(\frac{A_n}{A_1} \right)^2 \right) \quad (28)$$

$$\text{Define: } \delta = \sum_2^N n \left(\frac{A_n}{A_1} \right)^2 \quad (29)$$

$$(24), (29) \text{ into } (18) \rightarrow C_{D,i} = \frac{C_L^2}{\pi AR} (1 + \delta) \quad (30)$$

$$\text{Define Oswald's Efficiency Factor: } e = \frac{1}{1 + \delta} \quad (31)$$

$$(21) \text{ into } (20) \rightarrow C_{d,i} = \frac{C_L^2}{\pi e AR} \quad (32)$$

Note that equation (32) looks almost identical to equation (17), except the general wing case has Oswald's Efficiency Factor included in the equation. Given that δ is always non-negative, e is always less than or equal to 1. Any value for e less than 1 would result in a higher induced drag coefficient for a given lift coefficient. $e = 1$ in the elliptic case, which means that the elliptic wing distribution corresponds to the lowest induced drag coefficient for a given lift coefficient. This further reaffirms that the elliptic wing distribution is the ideal planform shape.

D. Results and Discussion

All of the calculations from this section of the report were conducted using Python in a Jupyter Notebook. The code is available to view in the appendix of this report.

1. Calculating Efficiency For Different Planforms

To instill confidence in the numerical approach, the efficiency factor and Fourier coefficients of an elliptic wing planform were calculated first. The results are given in the following tables:

Table 1 Elliptic Wing Efficiency Characteristics for Fixed Angle of Attack ($\alpha = 5^\circ$), Varying Aspect Ratio

Aspect Ratio	A_1	A_2	A_3	e
8	0.01047	1.15×10^{-19}	-3.25×10^{-20}	1.0
10	0.008727	6.73×10^{-20}	7.15×10^{-20}	1.0
12	0.00748	-1.24×10^{-20}	1.25×10^{-19}	1.0

Table 2 Elliptic Wing Efficiency Characteristics for Fixed Aspect Ratio (AR=8), Varying Angle of Attack

$\alpha(\text{deg})$	A_1	A_2	A_3	e
-4	-0.02094	-2.30×10^{-19}	6.50×10^{-20}	1.0
4	0.00698	7.67×10^{-20}	-2.18×10^{-20}	1.0
12	0.0349	3.84×10^{-19}	-1.08×10^{-19}	1.0

By inspecting the table, a few conclusions can be drawn about the elliptic wing distribution. The Fourier coefficients involved in the calculation change each time, whether aspect ratio or angle of attack is changed. It is important to note that only the first Fourier coefficient is non-zero. The second and third coefficients are so small that they are effectively zero. However, the efficiency factor for the elliptic distribution is always equal to one. In every scenario, an elliptic wing distribution will always be the most efficient.

The same calculations were run for a rectangular wing. The same values were used as with the elliptic wing

distribution to allow for more direct comparison. The results for the rectangular wing are given below in the following tables:

Table 3 Rectangular Wing Efficiency Characteristics for Fixed Angle of Attack ($\alpha = 5^\circ$), Varying Aspect Ratio

Aspect Ratio	A_1	A_2	A_3	e
8	0.0235	1.05×10^{-18}	0.00335	0.937
10	0.0196	-1.09×10^{-19}	0.00312	0.921
12	0.0168	-2.62×10^{-20}	0.00290	0.907

Table 4 Rectangular Wing Efficiency Characteristics for Fixed Aspect Ratio (AR=8), Varying Angle of Attack

α (deg)	A_1	A_2	A_3	e
-4	-0.00672	-6.12×10^{-19}	-0.00096	0.937
4	0.0202	1.17×10^{-18}	0.00287	0.937
12	0.0470	2.10×10^{-18}	0.00670	0.937

The rectangular wing calculations provide a bit more insight into how the model works. Like with the elliptic distribution, the Fourier coefficients always change for each calculation. But unlike the elliptic distribution, the Fourier coefficients are not all non-zero. In this case, the third Fourier coefficient is a little under a power of ten smaller than the first Fourier coefficient, which is by no means negligible. The efficiency factor of the rectangular wing distribution drops as a result.

Varying aspect ratio reveals a flaw in the Prandtl Lifting-Line Theory model as implemented in this report. As aspect ratio increases for a given planform, the efficiency of the wing should increase. This is because raising the aspect ratio brings the wing closer to approximating an infinite wing, which does not experience any induced drag. As the aspect ratio decreases, the model used in this report outputs an efficiency factor much higher than reality. Hence, it must be emphasized that Lifting-Line Theory provides the best accuracy for moderate aspect ratios (aspect ratios of around 8).

The efficiency factor does not change when varying the angle of attack for a fixed aspect ratio. This result makes physical sense because the efficiency factor of a wing is a characteristic of the wing geometry. Therefore, the efficiency factor should not change based on its orientation, which is validated in this model.

2. Lift and Induced Drag for Varying Aspect Ratio

One of the most important characteristics when designing a finite wing is aspect ratio. Aspect ratio is essentially a measure of how slender a wing is. Wings with high wingspans and short chord lengths have high aspect ratios. To

help inform the consultation for the preliminary wing design, it is important to model how varying aspect ratio affects lift generation compared to induced drag generation. Therefore, the lift coefficient versus induced drag coefficient was plotted for a given wing shape with varying aspect ratio. The resulting plots are called drag polars, and they help visualize how much lift a wing can generate for a certain amount of induced drag.

Drag polars were plotted for a rectangular wing. The plots were generated by looping the code multiple times for different wingspans stored in a list. The span dictated the aspect ratio because the chord length was held constant at one meter and the wing is rectangular, so the chord does not vary. The resulting curves are shown below:

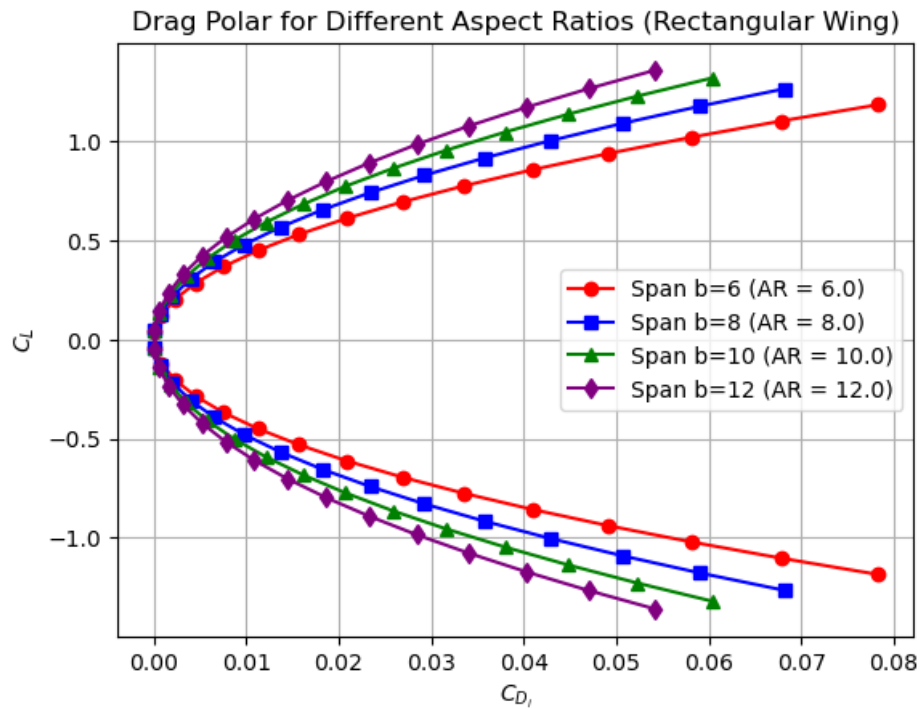


Fig. 12 Lift Coefficient vs. Induced Drag Coefficient for a Rectangular Wing with Varying Aspect Ratio

In Fig. 12, as aspect ratio increased, the drag polars got wider. This demonstrates that as aspect ratio increases, the wing can generate more lift with the same induced drag. This result corresponds to physical intuition as mentioned earlier in this section. Increasing aspect ratio moves the wing closer to approximating an infinite wing, corresponding to less induced drag. This result also reaffirms the effectiveness of the model used to calculate the efficiency factors, lift coefficients, and induced drag coefficients. Despite outputting worse efficiency factors as aspect ratio increased for a rectangular wing, the model still correctly outputs wider, more effective drag polars for higher aspect ratio wings.

3. Lift and Induced Drag for Different Wing Shapes

Given that Prandtl Lifting-Line Theory can only model straight wings (no sweep), the three wing shapes that shall arise in this section are elliptic, rectangular, and tapered wings. The wingspan of the plane is constrained to 12 meters

as was given in the problem statement. Therefore, it is important to evaluate the effectiveness of each wing shape for a given wingspan. Since efficiency increases with aspect ratio, and aspect ratio is a function of wingspan, then the new plane design shall have a wingspan as close to the constraint wingspan as possible. To add a tolerance, the different wing shapes were all modeled with a wingspan of 11 meters.

The lift coefficient versus induced drag coefficient were plotted for four wing shapes: elliptic planform, rectangular planform, tapered planform with a high taper ratio, and a tapered planform with a low taper ratio. The elliptic wing was modeled by its chord length having a sinusoidal distribution with respect to θ_j , the discretized points along the wing. The tapered wings were modeled by having its chord length vary between the defined tip chord length at the ends and defined root chord length in the center. The linear change between the tips and root was approximated with a sinusoidal function that depended on θ_j . The resulting curves are plotted below.

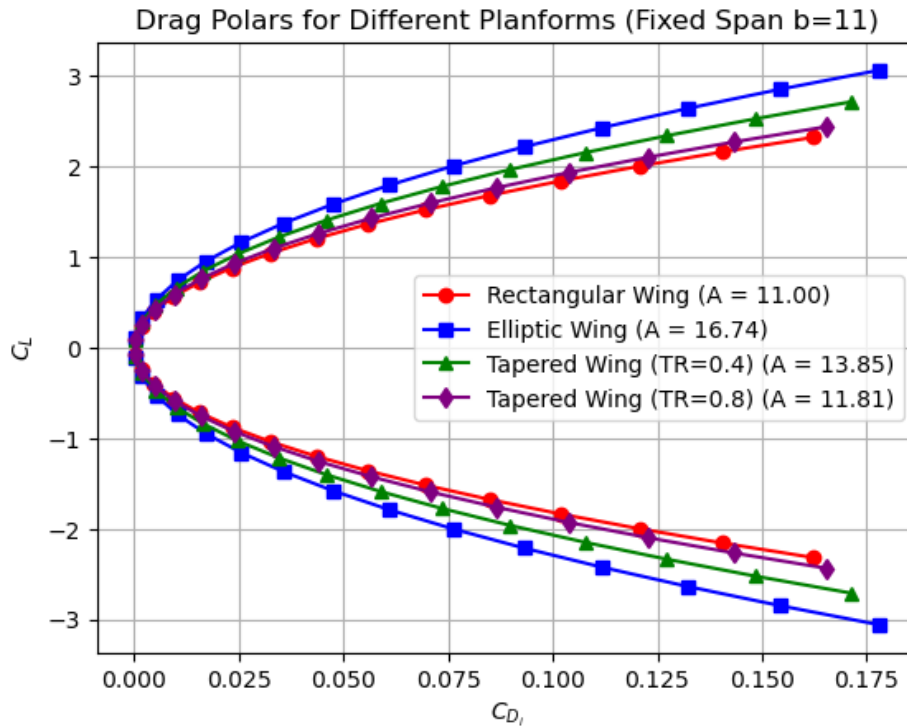


Fig. 13 Lift Coefficient vs. Induced Drag Coefficient for Different Wing Planforms at a Fixed Wingspan

Predictably, the elliptic distribution had the highest lift coefficient for a given induced drag coefficient. This is illustrated in the figure by a wider drag polar compared to the other polars. The rectangular planform had the narrowest drag polar, showing that it is the least effective at generating lift for a given wingspan. Therefore, a rectangular shape lends itself to generating more induced drag. This finding is corroborated by the drag polars for the two tapered wings. The wing with a higher taper ratio had a narrower drag polar than the wing with a lower taper ratio. A higher taper ratio means that the tip chord length is closer to the root chord length, which means that it more closely resembles a

rectangular planform compared to the low taper ratio wing.

Fig. 13 also displays the aspect ratio for each wing shape at a span of 11 meters. The elliptic wing distribution has the highest aspect ratio of the four wing shapes, which further suggests why it should be the most effective at generating lift for a given amount of induced drag.

4. The Effect of Geometric Twist on Lift and Induced Drag

Geometric twist describes the changing of a wing's geometric angle of attack spanwise along the wing with respect to the free stream velocity. It can be implemented in a wing to generate different lift distributions than a perfectly straight wing would. This is not to be confused with aerodynamic twist, which varies camber along the span of the wing.

Geometric twist is labeled either washout or wash-in. Washout means that the root of the wing has a higher angle of attack with respect to the free stream velocity than the tips. For wash-in, the opposite is true – the root of the wing has a lower angle of attack than the tips.

Three drag polars were created to evaluate the effects of geometric twist on a rectangular wing: one for a wing without twist, one with washout, and one with wash-in. Geometric twist was modeled in the code by varying the angle of attack based on what point on the wing was being evaluated in the loop. The angle of attack still varied in order to create the drag polars, but the variation of angle of attack also accounted for the discretization point (θ_j) was being evaluated in the loop. A negative cosine function of $2\theta_j$ was used to ensure that the maximum positive twist took place at the root, and the maximum negative twist took place at the tips. The resulting curves are plotted below.

Fig. 14 shows that small geometric twist angles only causes slight variations in the drag polars for a given wing. The wing with washout has a slightly greater lift coefficient for a given induced drag coefficient value, meaning that it is slightly more effective than the other twist configurations at generating lift. The opposite can be said for the wash-in wing. It is slightly less effective at generating lift than the wing with no twist. However, these changes in lift coefficient versus induced drag coefficient are rather miniscule.

E. Conclusion

According to Prandtl's Lifting-Line Theory as it was implemented in this report, it is essential that the company's wing design shall have as high of an aspect ratio as permissible. This will ensure a high aspect ratio without having to shorten the chord length. A shortened chord could introduce structural issues in the wing. Additionally, the wing shall be tapered with a relatively low taper ratio. This is supported in the analysis, as the low-taper-ratio wing came the closest to replicating the drag polar of the elliptic wing distribution. While washout twist did slightly improve the lift coefficient vs induced drag coefficient for the rectangular wing, this minor improvement does not justify the added manufacturing difficulty. Operating under the assumption that the trend shown in this report will persist for tapered wings, adding geometric twist to the wing is not justified.

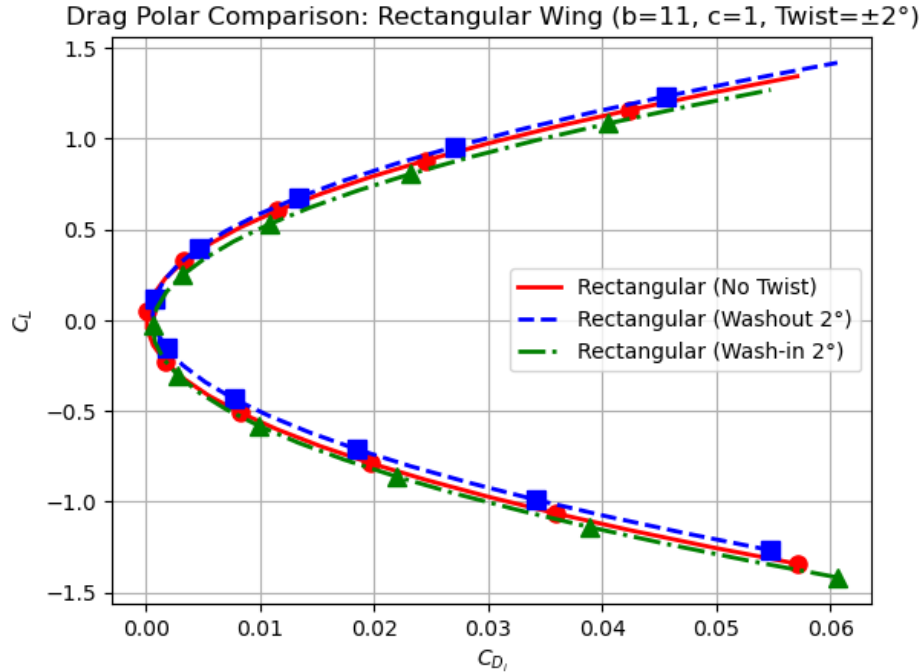


Fig. 14 Lift Coefficient vs. Induced Drag Coefficient for a Given Rectangular Wing with Varying Geometric Twist

Therefore, the official recommendation of this report is to design a wing with an 11 meter wingspan, 1 meter root chord length, 0.4 meter tip chord length, and no geometric twist. The wingspan is set at 11 meters to allow for a margin of error when transporting the plane in and out of the hangar. This wing is designed with a taper ratio of 0.4, which is low enough to approach elliptic wing levels of lift generation.

IV. Connection Between Real World Problem, Academic-Style Problem, and Course Content

The problems discussed are closely intertwined both with each other and with the course content. The real world problem explores the historical basis, trends, and major observed differences between wing shapes, while the academic-style problem takes a numerical and analytical approach to compare the same wing shapes. Both sections particularly address the effect of wing shape on induced drag and span efficiency factor. These sections also connect to course material via induced drag, since span efficiency and performance are intrinsically connected to this concept. The academic-style problem uses Prandtl lifting line theory as a basis for its model of span efficiency and induced drag. These concepts are further connected to a litany of concepts discussed in this course such as downwash, wingtip vortices, the Kutta Joukowski theorem, lift slope plots, drag polars, and many other core foundational concepts related to incompressible, inviscid flow.

The academic-style problem makes some key assumptions which distinguish it from the real world problem. The assumptions made include incompressible, steady, inviscid, irrotational flow, limited to straight wings at small angles

of attack and attached flow with no aerodynamic twist. These assumptions do not invalidate the findings, since the investigation seeks to make general observations applicable to flight under reasonable conditions, and the assumptions make the model feasible to implement. These assumptions do introduce some limitations, however. The real world problem particularly addresses a flaw of elliptic airfoil shapes: the stall behavior can be abrupt and unpredictable. The real world problem also discusses the benefit of stability of the rectangular wing planform. These findings come from real experimentation and experience with planes which have been flown and thus do not suffer from the same simplifications. Any assumption of attached flow will not predict stall. The computational model more specifically hones in on some particular performance metrics, providing a limited but valid conclusion for a subset of wing shapes.

V. Team Management

A. Tasks and Schedule

The schedule for this project was designed to be flexible. With many final projects, exams, and other obligations filling the group's time as the semester comes to a close, it was vital to allow my group to prioritize their most time-sensitive work first. As a result, the schedule we developed focused on key phases of work in which we could individually explore ideas and complete tasks, and then we could meet together to discuss progress, thoughts, and concerns. At each meeting, we discussed our purpose for the meeting, the events and progress achieved preceding the meeting, the goals for the meeting, the tasks to be completed for the next meeting, and the group's availability for the next meeting. Below, a calendar reflecting the schedule of meetings and outside obligations for the group can be found. This calendar was used by the group to determine meeting times and dates without conflicting with other obligations.

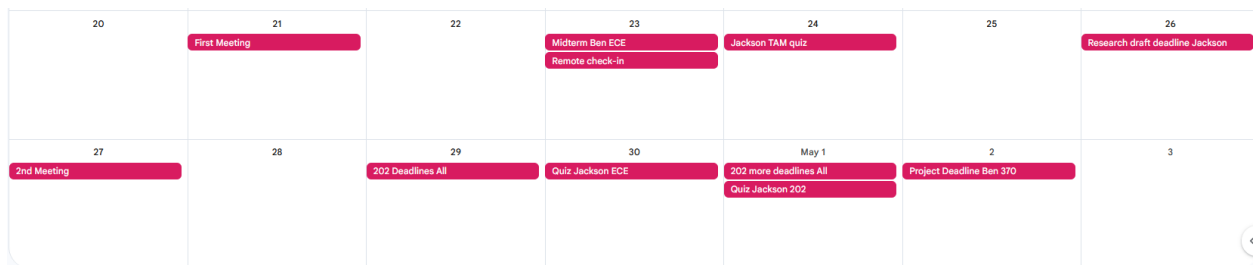


Fig. 15 The schedule of team meetings and major individual or shared obligations for the first two weeks of the project.



Fig. 16 The schedule of team meetings and major individual or shared obligations for the final week of the project.

To explicitly distribute the workload for the project, we largely relied on strategies employed in the previous projects, while incorporating lessons learned along the way. Ben McDonald, responsible for the academic-style problem, was tasked as the primary investigator to determine the overall problem we would work on. This entailed finding an interesting academic-style problem with some real-world analog or connection. Experience from the previous projects informed this decision that we as a group wanted the academic-problem role to be the primary concern. Next, Ben was responsible for writing his problem statement, writing any necessary code, and fleshing out a concrete mathematical basis for his investigation in the context of and extending material from the course. Finally, he was tasked with the results and discussion of his findings. Jackson Rees, responsible for the real-world problem, was tasked with finding a real world problem related to the academic-style problem which interested him and related to the course material. He communicated throughout preliminary investigations with Ben, allowing a dialogue to determine a mutual interest. Next, Jackson was responsible for surveying the research related to the real world problem, investigating both textual sources to form a basis of understanding on the topic, and then delving further into academic investigations to further develop a deep understanding of some of the nuances of the topic. Then he was responsible for writing a well-structured report exploring his findings and contextualizing his research in history, aviation, and with respect to the academic-style problem and the course. I, Max Lutter, was responsible for managing the team, structuring our schedule and tasks, and contributing to fulfill the workload of my group mates. I was tasked with formatting the mathematical symbols of the derivation into LaTeX, formatting the references for the real world problem, and with writing the team management section. Finally, all group members contributed to the connection section, providing insight into the connections between the real world problem to the academic-style problem and all of this to the course content.

B. Team Communication

The means of communication for our group was primarily through iMessage. We have maintained a group chat throughout the semester, allowing for consistent, clear communication. Additionally, we share coursework together including AE 311 and 202, providing daily opportunities for discussion. While this means of communication can be inconsistent because of its undocumented and casual nature, we made an effort as a group to send important details of our conversations through the iMessage group chat to serve as a reference later.

C. Challenges and Lessons Learned

The main challenge I encountered when managing this group was finding time when everyone was free to meet to work together. Since the end of the semester tends to be quite busy, it was challenging to find time to work together and even difficult to find time to work individually. As a result, it was important to keep the project at the front of our minds so that opportunities to make progress would not go to waste. If I were to go back I think I would make a greater effort

to make dedicated meetings for the project. The most productive times of work for our group came when we met in person to discuss the project and work together on our respective tasks. This would still be difficult to make happen, but I think I could have pushed harder for this to happen without putting unnecessary pressure on my group.

Another challenge we encountered as a group was to limit the scope and goals of the report. All of the members of the group want to put forth a solid report and are ambitious in this goal, but this can extend the amount of time it takes to complete a finished product. Towards the end of the project, there was some concern raised about finishing before the extended deadline. As a result, some triage calculation had to be made to prioritize the core objectives of the project and to recognize elements which are more supplemental. I think in future projects I would try to force my groupmates to define the core objectives or essentials of their research and investigations, allowing the team leader to manage the workload and better define a reasonable project scope to avoid a rush at the deadline.

VI. Conclusion

The real world problem investigates the effectiveness and evolution of different wing planforms, delving into historical applications and trends for wing shapes with various design objectives. This investigation found that an elliptic wing design was the most efficient design in terms of the span efficiency-factor, which dictates the lift a wing is able to generate while minimizing drag. While the elliptic wing design was theorized to be the most optimal shape, the investigation found some major limitations, such as abrupt and unpredictable stall behavior and immense manufacturing difficulty. The investigation further explores simpler planforms such as a rectangular wing shape, which offers greater stability and lower manufacturing cost, with the drawback of greater induced drag, and a tapered wing shape, which offers a healthy medium of low induced drag and high stability, while avoiding the heavy burden of manufacturing costs. The real world problem then defines key performance characteristics such as the span efficiency factor.

The academic-style problem sought to determine a viable design for a particular use. Their objective was to determine a wing shape which met the cost and performance requirements of their employer. In this pursuit, the academic-style problem first derives a mathematical model using finite wing theory to understand the performance of tapered wing design and elliptical wing design, specifically honing in on the effect of induced drag. This model is then used along with known performance metrics such as lift to drag ratios to code a tool able to determine the magnitude of efficiency lost when using a tapered wing design instead of an elliptical wing design, and to calculate the Fourier coefficients used to model the circulation distribution over the wing. This section also briefly discusses geometric twist, determining that any benefit it would have is marginal, especially in a case in which the manufacturing cost is a large concern.

These two problems both investigate the effect of wing shape on induced drag and wing performance, finding that the taper ratio or wing shape has a strong effect on the effect of induced drag and that a tapered wingtip shape benefits from an approximately elliptic lift distribution while massively saving manufacturing costs and complexity.

Both problems directly connect to course concepts related to induced drag, downwash, and finite wing theory. The academic investigation used finite wing theory, discussed at length in the course, to evaluate wing performance, while also determining evaluation criteria from course knowledge. The investigation. The academic problem uses Prandtl lifting line theory to derive its performance metrics and to model the performance of the different wing shapes.

The approach I took while managing this team took multiple phases. In the early stages of the project, my main priority was to facilitate a discussion and exploration of possible topics to explore. This consisted of frequent conversations and small chats to keep the group thinking and curious about the subject, asking questions and making suggestions to guide a preliminary investigation. I largely wanted to let my group decide for themselves what they might find interesting to explore, so I was careful to keep the pressure low, allowing them time to come to a comfortable conclusion while still emphasizing the project's objectives. Once we decided on a problem, my role shifted to more directly lower their respective workloads. By pitching in to help with more time intensive tasks such as digitizing plots for analysis, and referencing sources, I allowed their time to be spent on research and academic investigation. Simultaneously, I was able to still ask probing questions, provide feedback on their writing and presentation, and discuss a concrete plan of attack.

Appendix

A. Use of Artificial Intelligence

ChatGPT was used to aid the development of the computational model and for some $\text{\LaTeX}$ formatting.

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